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Competition and the Cost of Debt

Contents

1 Big picture

- **Question.** Does product market competition affect the price at which banks lend to firms?
- **Setting.** The paper studies U.S. syndicated bank loans from Dealscan merged with Compustat, CRSP, Hoberg–Phillips industry concentration measures, and trade data. The main sample covers 12,256 loans for 2,900 firms from 1992 to 2007.
- **Core challenge.** Competition may be endogenous to firms' financing choices. A simple positive correlation between loan spreads and competitive industries could reflect omitted risk, reverse causality, or firms strategically shaping industry structure.
- **Main economic idea.** Banks price debt mainly from *default risk* and *loss given default*. Competition can affect both: it can raise cash-flow risk and predation risk, and it can lower liquidation values by weakening asset redeployability.
- **Main hypotheses.**
  - More competition should raise loan spreads because banks perceive greater "competitive risk."
  - The effect should be especially strong when weak firms face strong rivals, when strategic interactions are intense, and when assets are hard to liquidate.
- **Main empirical design.** The paper first relates loan spreads to industry competition using a rich panel of loans and firm controls. It then addresses endogeneity with a quasi-natural experiment based on large import tariff reductions that intensify foreign competition.
- **Main baseline result.** Firms in competitive industries pay significantly higher loan spreads. In the simplest specification, the coefficient on `Competition` is 0.084, implying about an 8.4% higher spread for firms in the most competitive quartile. With the sample average spread of 178 basis points, that is about 15 basis points, or roughly $463,500 more per year on the average loan.
- **Main identification result.** Using large reductions in industry import tariffs, the paper shows that an exogenous increase in competitive pressure raises loan spreads by about 15% to 22% in the baseline tariff-shock designs.
- **Main heterogeneity result.** The effect of competition is much larger when small firms face large rivals, when industries are highly concentrated, and when assets are specific or illiquid.
- **Main contribution.** The paper connects product-market structure to debt pricing. It argues that banks do not only price standard default risk; they also price risk coming from strategic competition in the product market.
- **Economic takeaway.** Competition is not just about lower profits or lower markups. It also raises the external financing cost of firms, especially when the firm is strategically exposed and collateral is weak.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The paper studies bank loan pricing rather than bond pricing, so the dependent variable is observed directly in loan contracts.
- The sample is broad across industries, and the main competition proxy varies over time.
- The tariff-shock design later provides a quasi-experimental source of changes in competition.

Interpretation. The paper is built to say something about *cost of debt*, not just leverage or investment.

2.2 Sample construction

- The starting point is the CRSP–Compustat quarterly database merged with a July 2008 extract of Dealscan.
- The baseline sample covers 1992–2007.
- Loans missing borrower ID, pricing, maturity, or size are dropped.
- The final panel contains 12,256 loans for 2,900 distinct firms in 183 three-digit SIC industries.

Interpretation. The unit of observation is a loan, not a firm-year. That is why the paper can study loan-contract pricing directly.

2.3 Dependent variable: the cost of debt

The main dependent variable is the logarithm of the all-in-spread drawn from Dealscan. This is the amount the borrower pays in basis points over LIBOR or a LIBOR-equivalent benchmark for each dollar drawn, including annual fees paid to the lending group.

Important summary facts are:

- average loan spread: 178 basis points,
- average maturity: 44 months,
- average size: $310 million,
- average syndicate size: 8 banks,
- secured loans: 73% of the sample,
- dividend restrictions: 82%,
- average number of financial covenants: 2.82.

Interpretation. The paper studies economically large bank loans, not tiny private credit lines.

2.4 Main measure of product market competition

The baseline proxy is the fitted Herfindahl–Hirschman Index (HHI) at the three-digit SIC level from Hoberg and Phillips. Higher HHI means more concentration and therefore *less* competition.

The paper defines

$$Competition_{j,t} = \nVdash\{HHI_{j,t} \text{ is in the lowest quartile of the yearly distribution}\}. \tag{1}$$

This dummy is preferred in the baseline because it gives an intuitive interpretation and reduces sensitivity to noise in the exact HHI level.

The paper also uses alternative competition proxies:

- TNIC HHI and TNIC C4 from Hoberg–Phillips text-based industries,
- Compustat HHI and Compustat C4,
- import penetration,
- tariff-rate reductions as competitive shocks.

2.5 Borrower controls

The baseline regressions control for variables that proxy for default risk, financial distress, growth opportunities, and loan characteristics:

- log assets,
- market-to-book,
- leverage,
- tangibility,
- cash-flow volatility,
- default probability,
- profitability,
- loan size,
- log loan maturity,
- loan type dummies,
- and, in some specifications, year and industry fixed effects.

The default probability follows Bharath and Shumway's expected default frequency logic.

Interpretation. The paper tries hard to ensure that "competition" is not just a disguised proxy for obvious credit risk.

2.6 Competition-shock variable from tariffs

For manufacturing industries, the paper computes ad valorem import tariffs as customs duties divided by the free-on-board value of imports. It defines a large competitive shock when a tariff reduction in an industry is at least three times larger than that industry's median tariff-rate reduction.

This yields a post-shock indicator:

$$\textit{Post-reduction}_{j,t} = \mathbb{1}\{\text{industry } j \text{ has already experienced a large tariff cut by time } t\}. \tag{2}$$

Key facts:

- tariff data are available for 1992–2005,
- the matched tariff-loan sample has 5,331 loans for 1,372 firms in 96 three-digit SIC industries,
- 54 large tariff reductions are identified,
- 42 industries never experience such a shock,
- 30 of the 54 occur in 1995, around NAFTA, but shocks also occur later, reducing concern that the design is one-year specific.

2.7 Summary statistics that matter

Some important sample moments are:

- average book leverage: 0.31,
- average market-to-book: 1.41,
- average assets: $3.46 billion,
- mean HHI: 0.062,
- mean `Competition` dummy: 0.260,
- correlation between loan spreads and `Competition`: positive,
- correlation between loan spreads and HHI: negative.

Interpretation. Even before multivariate work, more competitive industries look more expensive to borrow in.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic framework

The paper starts from the idea that

$$\text{Cost of bank debt} = f(\text{Default risk, Loss given default}). \tag{3}$$

Competition can increase the first term by reducing profits and raising business risk, and it can worsen the second term by lowering asset liquidation values.

Interpretation. This compact expression is the conceptual map of the whole paper.

3.2 Baseline panel regression

The main loan-pricing equation is

$$y_{i,j,t} = \delta \, Competition_{j,t-1} + \boldsymbol{\beta}' X_{i,t-1} + \alpha_t + Z_j + \lambda_l + \varepsilon_{i,j,t}, \tag{4}$$

where:

- $y_{i,j,t}$ is the log loan spread,
- i indexes borrowers,
- j indexes industries,
- t indexes the quarter of loan issue,
- $X_{i,t-1}$ is the vector of firm and loan controls,
- α_t are year effects in some specifications,
- Z_j are three-digit SIC industry effects in some specifications,
- λ_l are loan-type effects.

The coefficient of interest is δ.

Interpretation. A positive δ means that competition raises the spread that banks demand even after conditioning on standard observables.

3.3 Alternative competition specifications

The paper re-estimates the same basic model with:

- TNIC-based competition measures,
- Compustat-based concentration measures,
- lender fixed effects,
- credit-rating fixed effects,
- governance controls such as the G-index,
- and Fama–MacBeth or between estimators for cross-industry evidence.

Interpretation. This is not one fragile specification. The same economic message shows up under many ways of coding competition.

3.4 Quasi-experimental tariff-shock model

To address endogeneity, the paper estimates

$$y_{i,j,t} = \lambda \, Post\text{-}reduction_{j,t} + \boldsymbol{\beta}' X_{i,t-1} + Z_j + \lambda_l + \varepsilon_{i,j,t}. \tag{5}$$

Again, the dependent variable is the log loan spread. The coefficient of interest is now λ, the effect of an exogenous increase in competition caused by a large tariff cut.

Interpretation. This is essentially a difference-in-differences design where industries that have not yet been hit by a tariff shock serve as controls for those that have.

3.5 Dynamic event-timing test

To study pre-trends and persistence, the paper replaces `Post-reduction` with timing indicators:

$$y_{i,j,t} = \theta_{-1}\, \mathit{Before1}_{j,t} + \theta_0\, \mathit{Before0}_{j,t} + \theta_{+1}\, \mathit{After1}_{j,t} + \theta_{+2}\, \mathit{After2+}_{j,t} + \boldsymbol{\beta}' X_{i,t-1} + \cdots \tag{6}$$

where:

- `Before1`: one year before the tariff cut,
- `Before0`: the year of the cut,
- `After1`: one year after,
- `After2+`: at least two years after.

Interpretation. If pre-shock coefficients are small while post-shock coefficients are positive, the timing is consistent with tariffs causing spreads rather than spreads predicting tariffs.

3.6 Change regression

The paper also estimates a first-difference specification using firms with comparable loan types before and after a shock:

$$\Delta \log(\text{loan spread}) = \kappa\, \Delta(\text{Tariff rate}) + \boldsymbol{\rho}' \Delta X + u. \tag{7}$$

Interpretation. This design removes time-invariant firm-level omitted factors and checks whether spread changes move mechanically with tariff changes.

3.7 Cross-sectional heterogeneity models

The baseline competition equation is re-estimated separately across subsamples defined by:

- rival size and rival credit quality,
- industry concentration and similarity of operations,
- asset specificity and asset illiquidity,
- number of potential buyers.

Interpretation. These exercises try to say *what kind* of competition risk is priced by banks.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

The baseline coefficient on `Competition` is identified from cross-industry and, with industry fixed effects, from within-industry changes in concentration over time. The rich control set absorbs many standard determinants of credit spreads.

Why this helps. If competition just stood in for leverage, default risk, or growth opportunities, the coefficient should disappear once those controls are added. It does not.

4.2 Why reverse causality is a real concern

Competition is not automatically exogenous. Firms may use financing policy strategically to shape product-market outcomes. The paper explicitly acknowledges this concern, citing theories in which capital structure affects rivalry.

Implication. Pure cross-sectional evidence cannot by itself establish causality.

4.3 Why the tariff design is credible

The tariff-shock design is credible for two reasons:

- tariff reductions bring genuine real-side increases in foreign competition,
- and the timing appears at least partly unanticipated from the perspective of firms' financing choices.

Supporting facts in the paper are:

- average tariffs fall from about 3% before a shock to below 1.5% after it,
- import penetration rises from 19.5% to 24.1%,
- firms in affected and unaffected industries look similar in financing policies before the shock,
- and there is no obvious reason industries would lobby *for* lower tariffs just to alter borrowing costs.

4.4 Why pre-trends matter

The dynamic specification is crucial. If spreads were already rising before the tariff cut, the story would look much more like reverse causality or omitted trends.

What the paper finds instead:

- `Before1` and `Before0` are insignificant,
- `After1` is positive and significant,
- `After2+` is even larger.

Interpretation. The timing is consistent with competition increasing spreads after the shock rather than before it.

4.5 How the paper handles sample-selection concerns

A possible objection is that highly exposed firms may simply stop borrowing after competition intensifies, biasing the sample.

The paper addresses this in three ways:

- firm fixed effects,
- restricting to refinancing loans,
- restricting to firms observed borrowing both before and after the shock.

Interpretation. The effect survives in all three cases, so it is not just a "new-loan composition" artifact.

4.6 Why default risk is not a sufficient statistic

Even after controlling for leverage, cash-flow volatility, default probability, Z-score, and credit ratings, competition still matters.

Interpretation. Banks seem to price something broader than standard default risk—what the paper calls *competitive risk*. That broader concept includes predation risk and collateral-value risk.

4.7 Alternative explanations the paper addresses

The paper rules out or weakens several alternatives:

- lender-specific pricing with lender fixed effects,
- broad governance differences with the G-index,
- alternative industry definitions and competition measures,
- omitted time-invariant firm characteristics with firm fixed effects and change regressions,
- and macroeconomic cycles with credit spreads, term spreads, real GDP growth, and a late-1990s dummy.

Interpretation. The empirical story is not hanging on one narrow definition of competition or one narrow credit-spread model.

4.8 What the paper can and cannot distinguish

The paper is very persuasive on the claim that competition raises loan spreads. It is weaker on pinning down the exact mix of mechanisms. In practice, the evidence is consistent with three channels at once:

- higher default risk,
- stronger predatory pressure from rivals,
- lower liquidation value when assets are illiquid.

5 Tables and figures

5.1 Table 1: summary statistics

Read:

- Mean loan spread is 178 basis points over LIBOR.
- Average loan maturity is 44 months and average loan size is $310 million.
- The average HHI is 0.062 and the mean competition dummy is 0.260.
- Loan spreads are positively correlated with the competition dummy and negatively correlated with HHI.

Economic message. Competitive industries are already more expensive to borrow in at the raw descriptive level.

Table 1
Summary statistics.
This table presents summary statistics for loans, borrowers, and for proxies of product market competition. Panel A presents means, medians, and standard deviations of loan characteristics. Panel B shows summary statistics of borrower characteristics. Panel C presents summary statistics of competition proxies. Finally, Panel D shows the correlation coefficients between loan spreads and the competition proxies. The sample period is from 1992 to 2007. All variables are defined in Appendix A. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

Panel A: Loan characteristics

	N	Mean	Median	SD	Min	Max
Loan spread (basis points)	12,256	178.38	150	127.16	17.5	580
Loan maturity (months)	12,256	44.04	45	23.82	11	121
Loan size (million USD)	12,256	310	115	509	0.85	2800
Loan size (to total assets)	12,256	0.18	0.13	0.17	0.01	0.98
Syndicate size	12,251	8.05	5	8.72	1	45
Secured dummy	8,638	0.73	1	0.44	0	1
Dividend restriction dummy	7,815	0.82	1	0.38	0	1
No. of financial covenants	7,525	2.82	3	1.19	1	7

Panel B: Borrower characteristics

	N	Mean	Median	SD	Min	Max
Total assets (billion USD)	12,256	3.46	0.71	6.89	0.00	34.46
Leverage	12,256	0.31	0.29	0.2	0	0.96
Profitability	12,256	0.03	0.03	0.03	−0.18	0.13
Market-to-book	12,256	1.41	1.13	1.01	0.32	8.44
Tangibility	12,256	0.34	0.28	0.24	0.01	0.9
Cash flow volatility	12,256	0.02	0.01	0.02	0	0.15
Default probability	12,256	0.17	0	0.31	0	1

Panel C: Product market competition proxies

	N	Mean	Median	SD	Min	Max
HHI	12,256	0.062	0.053	0.028	0.032	0.224
Competition	12,256	0.260	0	0.439	0	1
Compustat HHI	12,256	0.161	0.126	0.136	0	0.937
TNIC HHI	9,301	0.131	0.092	0.118	0	0.939
ΔTariff	5,331	−0.133	−0.062	0.361	−4.057	2.929

Panel D: Correlation coefficients

	Log(loan spread)	*Fitted HHI*	*Competition*	*Compustat HHI*	*TNIC HHI*	*ΔTariff*
Log(loan spread)	1.00					
HHI	−0.106***	1.00				
Competition	0.096***	−0.476***	1.00			
Compustat HHI	−0.051***	0.391***	−0.316***	1.00		
TNIC HHI	0.052***	−0.071***	0.046***	0.025**	1.00	
ΔTariff	−0.057***	0.129***	−0.065***	0.051***	0.021	1.00

Table 2
Loan spreads and firm characteristics across quartiles of HHI and firm size.
This table presents average loan spreads and firm characteristics for quartile portfolios of firms formed using the Herfindahl-Hirschman Index (HHI) at the three-digit SIC code industry level from Hoberg and Phillips (2010a). This HHI combines Compustat data with Herfindahl data from the US Commerce Department and employee data from the Bureau of Labor Statistics, covers private and public firms, and all industries. Higher values of the HHI are associated with lower levels of competition. In Panel A, I sort firms each calendar year based on the HHI and assign the firms into quartiles. Next, I compute average and median loan spreads for each quartile. The last column reports the difference in means and medians between competitive (*Q*1) and concentrated (*Q*4) industries. In Panel B, I additionally split firms into small and large firms within each HHI quartile based on total assets (below or above median, respectively) and report the averages of proxies for financial, default, and cash flow risk. I compute the statistical significance of the difference in means with a mean comparison *t*-test, and the difference in medians with a Wilcoxon rank-sum test. The sample period is from 1992 to 2007. All variables are defined in Appendix A. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

Panel A: Loan spreads for quartile portfolios based on the HHI

	Q1	Q2	Q3	Q4	Q1–Q4
Average loan spread	194.09	187.03	169.31	162.28	31.81***
Median loan spread	175.00	175.00	150.00	132.50	42.50***
Average number of loans	3,188	3,017	3,016	3,035	

Panel B: Loan spreads and firm characteristics for quartile portfolios based on the HHI and total assets

Small firms	Q1	Q2	Q3	Q4	Q1–Q4
Loan spread	253.85	240.78	228.49	211.92	41.93***
Leverage	0.307	0.332	0.335	0.338	−0.031***
Default probability	0.206	0.211	0.210	0.245	−0.039***
Cash flow volatility	0.027	0.024	0.022	0.021	0.006***
Large firms	Q1	Q2	Q3	Q4	Q1–Q4
Loan spread	134.32	133.32	110.13	112.47	21.85***
Leverage	0.349	0.367	0.334	0.331	0.018***
Default probability	0.113	0.187	0.136	0.145	−0.032***
Cash flow volatility	0.014	0.011	0.013	0.012	0.002***

5.2 Table 2: univariate evidence across competition quartiles

Read:

- Average spread in the most competitive quartile is 194.09 basis points versus 162.28 in the least competitive quartile, a difference of 31.81 basis points.
- The median spread difference is even larger: 42.50 basis points.
- Among small firms, the spread gap between competitive and concentrated industries is 41.93 basis points.
- Among large firms, the same gap is 21.85 basis points.

Economic message. Before any controls, competition is associated with noticeably more expensive debt, and the gap is especially large for small firms.

5.3 Table 3: baseline multivariate loan-spread results

Read:

- Without fixed effects, the coefficient on `Competition` is 0.084.
- With loan-type and year fixed effects, it is 0.067.
- With industry fixed effects—the baseline specification—it rises to 0.096.
- With credit-rating fixed effects, it is 0.094.
- With lender fixed effects, it is 0.082.
- With extra macro and firm controls, it remains positive at about 0.079.
- Fama–MacBeth and between-industry estimates are also positive.

Economic message. The result is not fragile. Competition raises spreads across a wide range of specifications.

Table 3
Competition and the cost of debt: panel results.

This table presents coefficient estimates of regressions which examine the effect of competition on loan spreads (Eq. (1)). The dependent variable is the logarithm of loan spreads. *Competition* is a dummy variable equal to one if the HHI at the three-digit SIC code industry level is in the lowest quartile of the yearly sample distribution, and zero otherwise. Column 1 presents the results without any fixed effects. Column 2 includes loan type and year fixed effects, and column 3 includes industry fixed effects at the three-digit SIC code industry level. Column 4 includes credit rating fixed effects, and column 5 lender fixed effects. In column 6, I include additional control variables. Column 7 controls for governance using the G-Index. In columns 8 and 9, I estimate Eq. (1) using a Fama and MacBeth (1973) and between industry estimator, respectively. Finally, column 10 directly includes the HHI as a proxy for competition. Lower values of the HHI indicate more competitive industries. I measure all independent variables as of the quarter prior to the loan start date. The sample period is from 1992 to 2007. All variables are defined in Appendix A. I report t-statistics using standard errors adjusted for within-firm clustering (OLS) or for one lag serial correlation (Fama and MacBeth) in parentheses below the coefficient estimates. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

Variables	*No fixed effects* (1)	*Loan type and year* (2)	*Baseline (SIC3)* (3)	*Credit Ratings* (4)	*Lenders* (5)	*Other controls* (6)	*G Index* (7)	*Fama-MacBeth* (8)	*Between regression* (9)	*HHI* (10)
Competition	0.084***	0.067***	0.096***	0.094***	0.082**	0.079**	0.109**	0.068***	0.070**	
	(3.62)	(3.46)	(3.00)	(2.94)	(2.52)	(2.45)	(2.07)	(4.20)	(1.96)	
HHI										−0.602**
										(−2.54)
Log(total assets)	−0.280***	−0.268***	−0.284***	−0.228***	−0.271***	−0.247***	−0.332***	−0.275***	−0.107***	−0.273***
	(41.91)	(39.58)	(43.76)	(34.57)	(37.17)	(32.51)	(28.92)	(27.86)	(4.25)	(27.74)
Market-to-book	−0.095***	−0.095***	−0.081***	−0.065***	−0.083***	−0.093***	−0.109***	−0.124***	−0.089**	0.036**
	(7.99)	(9.42)	(8.66)	(7.59)	(9.00)	(8.01)	(6.53)	(8.91)	(2.43)	(2.48)
Leverage	0.859***	0.734***	0.712***	0.517***	0.706***	0.761***	0.801***	0.833***	0.500*	0.567***
	(14.25)	(14.17)	(13.83)	(10.70)	(13.53)	(12.00)	(9.89)	(16.92)	(1.82)	(11.64)
Tangibility	−0.070	0.030	−0.298***	−0.205***	−0.295***	−0.356***	−0.190*	0.025	0.064	0.068**
	(1.50)	(0.75)	(4.88)	(3.91)	(4.88)	(5.84)	(1.79)	(0.81)	(0.77)	(2.31)
Cash flow volatility	0.575	0.558	0.186	0.779**	0.044	0.322	1.871**	1.131***	0.432	1.290***
	(1.17)	(1.38)	(0.47)	(2.07)	(0.11)	(0.79)	(2.40)	(2.95)	(0.24)	(3.76)
Default probability	0.458***	0.363***	0.356***	0.333***	0.316***	0.394***	0.587***	0.342***	0.359**	0.051*
	(11.33)	(10.09)	(9.67)	(10.73)	(8.50)	(10.80)	(11.01)	(9.59)	(2.01)	(1.73)
Profitability	−3.337***	−2.899***	−2.660***	−2.542***	−2.047***	−2.936***	−3.329***	−3.363***	−5.181***	−2.854***
	(10.96)	(10.93)	(10.99)	(11.19)	(8.20)	(11.84)	(5.63)	(12.81)	(3.20)	(10.82)
Loan size	−0.552***	−0.332***	−0.405***	−0.376***	−0.359***	−0.418***	−0.407***	−0.268***	0.893**	−0.275***
	(8.77)	(5.35)	(5.73)	(5.87)	(5.25)	(5.92)	(5.32)	(5.66)	(2.22)	(5.66)
Log(loan maturity)	0.141***	−0.065***	−0.049***	−0.067***	−0.050***	−0.063***	−0.059**	−0.019	−0.639***	−0.055***
	(11.09)	(4.20)	(3.26)	(4.67)	(3.23)	(4.03)	(2.17)	(0.94)	(4.42)	(3.16)
Credit spread						0.493***				
						(13.99)				
Term spread						0.010*				
						(1.78)				
Market share						−0.531***				
						(4.90)				
Stock return						0.001***				
						(3.18)				
Z-score						0.011***				
						(2.63)				
G-Index							−0.011**			
							(2.04)			
Loan type fixed effects	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	No	Yes	Yes	Yes	Yes	No	Yes	No	No	No
Industry fixed effects	No	No	Yes	Yes	Yes	Yes	Yes	No	No	No
Estimation method	OLS	OLS	OLS	OLS	OLS	OLS	OLS	Fama MacBeth	Between regression	Fama MacBeth
Observations	12,256	12,256	12,256	12,256	12,251	11,171	5,868	12,256	12,256	12,256
Adjusted/average/between R^2	0.55	0.64	0.68	0.73	0.70	0.67	0.70	0.66	0.78	0.70

Table 4
Competition and the cost of debt: alternative industry definitions and proxies for competition.

This table presents coefficient estimates of regressions which examine the effect of competition on loan spreads (Eq. (1)). The dependent variable is the logarithm of loan spreads. *Competition* is a dummy variable equal to one if the HHI is in the lowest quartile of the yearly sample distribution, and zero otherwise. Columns 1 and 2 present the results using the HHI based on the text-based network industry classification (TNIC) following Hoberg and Phillips (2011). Columns 3 and 4 show the results using the C4-Index based on the TNIC. Columns 5 and 6 present the results using the HHI based on Compustat data; and columns 7 and 8 show the results using the C4-Index based on Compustat data. I measure all independent variables as of the quarter prior to the loan start date. The sample period is from 1996 to 2007 for columns 1–4, and 1992–2007 for columns 5–8. All variables are defined in Appendix A. I report t-statistics using standard errors adjusted for within-firm clustering (OLS) or for one lag serial correlation (Fama and MacBeth) in parentheses below the coefficient estimates. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

	Competition (TNIC)		*C4-Index (TNIC)*		*Competition (Compustat)*		*C4-Index (Compustat)*	
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Competition (TNIC)	0.117***	0.030*						
	(4.65)	(1.92)						
C4-Index (TNIC)			−0.442***	−0.271***				
			(9.02)	(4.57)				
Competition (Compustat)					0.059***	0.042**		
					(2.67)	(2.26)		
C4-Index (Compustat)							−0.218***	−0.102
							(5.76)	(0.87)
Log(total assets)	−0.278***	−0.275***	−0.282***	−0.278***	−0.278***	−0.286***	−0.276***	−0.286***
	(22.82)	(67.77)	(22.99)	(36.80)	(28.21)	(81.76)	(27.51)	(42.93)
Market-to-book	−0.133***	−0.076***	−0.128***	−0.075***	−0.128***	−0.081***	−0.128***	−0.079***
	(7.37)	(12.27)	(7.17)	(7.22)	(9.01)	(14.83)	(9.02)	(8.17)
Leverage	0.661***	0.739***	0.724***	0.681***	0.765***	0.792***	0.767***	0.801***
	(11.47)	(20.47)	(11.65)	(11.84)	(15.72)	(25.42)	(15.62)	(14.35)
Tangibility	−0.069**	−0.374***	−0.110***	−0.363***	0.060**	−0.338***	0.047	−0.329***
	(2.26)	(9.05)	(3.89)	(5.24)	(2.02)	(9.43)	(1.37)	(4.89)
Cash flow volatility	1.872**	0.563*	1.215***	0.491	1.091***	0.247	1.108***	0.184
	(2.44)	(1.75)	(2.65)	(1.06)	(2.92)	(0.89)	(2.76)	(0.44)
Default probability	0.397***	0.390***	0.342***	0.387***	0.349***	0.351***	0.336***	0.345***
	(8.44)	(16.78)	(8.96)	(9.57)	(9.72)	(17.80)	(8.89)	(8.46)
Profitability	−3.446***	−2.782***	−3.472***	−2.612***	−3.194***	−2.753***	−3.380***	−2.731***
	(11.04)	(14.70)	(11.45)	(9.80)	(12.86)	(17.08)	(12.68)	(10.59)
Loan size	−0.385***	−0.376***	−0.388***	−0.426***	−0.312***	−0.376***	−0.305***	−0.371***
	(7.36)	(9.97)	(7.22)	(5.39)	(6.40)	(11.41)	(5.77)	(4.87)
Log(loan maturity)	−0.011	−0.009	−0.026	−0.026	−0.059***	−0.031***	−0.059***	−0.034***
	(0.45)	(0.67)	(1.17)	(1.51)	(3.23)	(2.88)	(3.01)	(2.31)
Loan type fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
Industry fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
Estimation method	Fama MacBeth	OLS	Fama MacBeth	OLS	Fama MacBeth	OLS	Fama MacBeth	OLS
Observations	9,264	9,264	9,164	9,164	12,256	12,256	11,824	11,824
Average/adjusted R^2	0.71	0.67	0.71	0.69	0.69	0.66	0.70	0.66

5.4 Table 4: alternative competition measures

Read:

- TNIC-based competition is also positively related to loan spreads.
- TNIC C4 is negatively related to spreads, meaning that more concentrated TNIC industries are cheaper to borrow in.
- Compustat HHI and Compustat C4 tell the same story.

Economic message. The result does not depend on one peculiar industry-classification scheme.

5.5 Why Table 4 matters conceptually

This table is important because competition is a difficult object to measure. The fact that the same pattern shows up under fixed SIC industries, text-based network industries, HHI measures, and C4 measures makes the interpretation much stronger.

5.6 Figure 1: timing of tariff reductions

Read:

- The figure plots the number of large tariff reductions over time from 1992 to 2005.
- Thirty of the 54 shocks occur in 1995, but there are additional shocks later in the sample.

Economic message. The tariff design is not driven by a single event date only, even though NAFTA clearly matters.

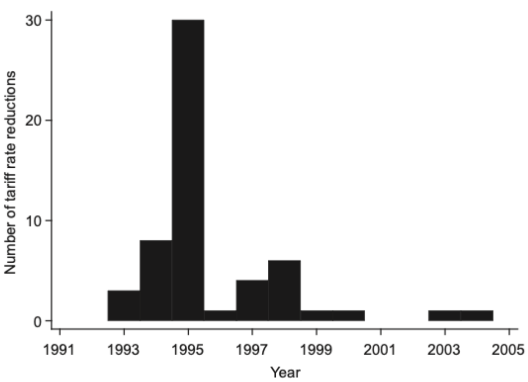

Fig. 1. Distribution of tariff rate reductions through time. This figure shows the number of tariff rate reductions for each year during the sample period 1992–2005. Tariff rates are computed at the three-digit SIC code industry level as duties collected at U.S. Customs divided by the Free-On-Board custom value of imports. An industry experiences a tariff rate reduction if the reduction is at least three times larger than the median tariff rate reduction in that industry.

Table 5
Loan spreads and firm characteristics before and after a reduction of import tariff rates.

This table presents average loan spreads and firm characteristics *before* and *after* a *large* reduction of import tariff rates for small and large firms. The tariff rate reduction is *large* if it is at least three times larger than the median tariff rate reduction in that industry. Panel A presents descriptive statistics for firms with a total asset size below the sample median, and Panel B shows the statistics for firms with a total asset size above the sample median. The sample period is from 1992 to 2005. All variables are defined in Appendix A. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

Panel A: Small firms			
	Before	*After*	*Difference*
Loan spread	222.54	244.15	21.61***
Default probability	0.169	0.199	0.031**
Cash flow volatility	0.024	0.026	0.002*
Leverage	0.290	0.289	−0.001
Panel B: Large firms			
	Before	*After*	*Difference*
Loan spread	88.30	128.12	39.81***
Default probability	0.076	0.133	0.057***
Cash flow volatility	0.013	0.013	0.000
Leverage	0.302	0.329	0.027***

5.7 Table 5: before and after tariff shocks

Read:

- For small firms, average spread rises from 222.54 to 244.15 basis points after a tariff shock, a 21.61-basis-point increase.
- For large firms, spread rises from 88.30 to 128.12 basis points, a 39.81-basis-point increase.
- Default probability rises for both groups.
- Cash-flow volatility rises for small firms, while leverage rises for large firms.

Economic message. The raw before/after comparison already points in the same direction as the main regression evidence.

5.8 Table 6: multivariate tariff-shock estimates

Read:

- Without fixed effects, `Post-reduction` is 0.169.
- With loan-type and industry fixed effects, it is 0.221.
- Using alternative shock definitions, the coefficient remains around 0.208 to 0.221.

Economic message. A large tariff cut—interpreted as an exogenous increase in product-market competition—raises loan spreads materially.

Table 6
Reductions of import tariff rates and the cost of debt.

This table presents coefficient estimates of the effect of tariff rate reductions on loan spreads. The dependent variable is the logarithm of loan spreads. In columns 1 and 2, *Post-reduction*$_{j,t}$ equals one if industry *j* has experienced a tariff rate reduction by time *t* that is larger than *three* times the median tariff rate reduction in that industry, and zero otherwise. In column 3, *Post-reduction*$_{j,t}$ equals one if industry *j* has experienced a tariff rate reduction by time *t* that is larger than *two* times the median tariff rate reduction in that industry, and zero otherwise. In column 4, *Post-reduction*$_{j,t}$ equals one if industry *j* has experienced a tariff rate reduction by time *t* that is larger than *three* times the *average* tariff rate reduction in that industry, and zero otherwise. Columns 2–4 include loan type and three-digit SIC code industry fixed effects. The sample period is from 1992 to 2005. All variables are defined in Appendix A. I report *t*-statistics using standard errors adjusted for within-firm clustering in parentheses below the coefficient estimates. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

	No fixed effects (1)	*Baseline* (2)	*abs(ΔTariff) > 2 × median* (3)	*abs(ΔTariff) > 3 × mean* (4)
Post-reduction$_{j,t}$	0.169***	0.221***	0.221***	0.208***
	(5.16)	(7.05)	(7.75)	(5.17)
Log(total assets)	−0.290***	−0.266***	−0.268***	−0.263***
	(31.159)	(30.33)	(30.45)	(30.13)
Market-to-book	−0.106***	−0.083***	−0.081***	−0.082***
	(6.74)	(6.90)	(6.79)	(7.08)
Leverage	0.760***	0.639***	0.646***	0.636***
	(8.28)	(8.10)	(8.17)	(8.02)
Tangibility	−0.312***	−0.426***	−0.439***	−0.445***
	(3.18)	(4.48)	(4.65)	(4.70)
Cash flow volatility	−0.580	−0.364	−0.425	−0.291
	(0.73)	(0.58)	(0.67)	(0.46)
Default probability	0.569***	0.492***	0.485***	0.494***
	(10.51)	(10.30)	(10.08)	(10.21)
Profitability	−3.579***	−2.896***	−2.877***	−2.986***
	(8.75)	(8.19)	(8.14)	(8.68)
Loan size	−0.604***	−0.401***	−0.403***	−0.388***
	(8.27)	(5.56)	(5.61)	(5.41)
Log(loan maturity)	0.139***	−0.087***	−0.086***	−0.087***
	(7.97)	(3.91)	(3.87)	(3.91)
Loan type fixed effects	No	Yes	Yes	Yes
Industry fixed effects	No	Yes	Yes	Yes
Observations	5,331	5,331	5,331	5,331
Adjusted R^2	0.59	0.68	0.68	0.68

Table 7
Reductions of import tariff rates and the cost of debt: additional tests.

This table presents coefficient estimates of the effect of tariff rate reductions on loan spreads. The dependent variable is the logarithm of loan spreads. *Post-reduction*$_{j,t}$ equals one if industry *j* has experienced a tariff rate reduction by time *t* that is larger than *three* times the median tariff rate reduction in that industry, and zero otherwise. Column 1 includes additional control variables compared to the baseline specification (column 2, Table 6). *I (1995 < year < 2001)* is a dummy variable equal to one if the year is between 1996 and 2000. Column 2 shows results including firm fixed effects (FE). Column 3 includes firm and year fixed effects. Column 4 shows results for loans that are specifically *refinancing* loans as defined by LPC's Dealscan database. Column 5 shows results for firms that took out at least one loan before and one loan after the tariff rate reduction. Column 6 shows dynamic effects and replaces *Post-reduction*$_{j,t}$ with four dummy variables: *Before*$^{-1}$ is a dummy variable equal to one for an industry that will experience a tariff rate reduction one year prior to the event, *Before*^{0} is a dummy variable equal to one for an industry experiencing a tariff rate reduction in that year, *After*^{1} is a dummy variable equal to one for an industry that experienced a tariff rate reduction last year, and *After*$^{2+}$ is a dummy variable equal to one for an industry that experienced the tariff rate reduction at least two years ago. Column 7 shows results using import penetration as a measure of competition from foreign firms. The sample period is from 1992 to 2005. All variables are defined in Appendix A. I report *t*-statistics using standard errors adjusted for within-firm clustering in parentheses below the coefficient estimates. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

Variables	*Controls* (1)	*Firm-FE* (2)	*Year-FE* (3)	*Refinance* (4)	*Before/After* (5)	*Dynamic* (6)	*Import* (7)
Post-reduction$_{j,t}$	0.154***	0.183***	0.080***	0.245***	0.231***		
	(4.79)	(3.51)	(2.72)	(5.21)	(5.55)		
Before^{-1}						0.115	
						(1.41)	
Before^{0}						0.109	
						(1.06)	
After^{1}						0.220**	
						(2.29)	
After^{2+}						0.349***	
						(4.23)	
Import penetration							0.768***
							(4.72)
Log(total assets)	−0.273***	−0.174***	−0.291***	−0.254***	−0.296***	−0.261***	−0.266***
	(30.97)	(5.02)	(15.29)	(−21.13)	(18.78)	(19.84)	(31.41)
Market-to-book	−0.076***	−0.082***	−0.075***	−0.093***	−0.097***	−0.078***	−0.082***
	(6.18)	(3.45)	(5.99)	(5.73)	(3.05)	(4.86)	(7.12)
Leverage	0.661***	0.567***	0.558***	0.748***	0.823***	0.631***	0.651***
	(8.52)	(4.34)	(7.97)	(7.38)	(5.19)	(6.89)	(8.78)
Tangibility	−0.393***	−0.724***	−0.281***	−0.434***	−0.687***	−0.447***	−0.413***
	(4.15)	(3.05)	(2.21)	(3.57)	(3.50)	(5.03)	(4.51)
Cash flow volatility	−0.204	0.896	0.516	0.102	1.308	−0.512	−0.048
	(0.33)	(0.75)	(0.76)	(0.11)	(0.97)	(0.79)	(0.08)
Default probability	0.470***	0.452***	0.344***	0.460***	0.529***	0.482***	0.503***
	(9.53)	(6.93)	(9.41)	(7.52)	(5.67)	(10.06)	(11.12)
Profitability	−2.815***	−2.743***	−2.114***	−3.316***	−2.638***	−2.887***	−2.900***
	(7.88)	(4.65)	(6.51)	(6.40)	(3.53)	(5.85)	(8.61)
Loan size	−0.351***	−0.447***	−0.357***	−0.464***	−0.294**	−0.411***	−0.389***
	(4.94)	(4.56)	(6.58)	(4.61)	(2.08)	(5.16)	(5.64)
Log(loan maturity)	−0.056**	−0.071***	−0.040**	−0.077**	−0.079*	−0.078***	−0.079***
	(2.55)	(2.63)	(2.35)	(2.36)	(1.93)	(4.12)	(3.68)
Credit spread	0.425***						
	(7.72)						
Term spread	−0.009						
	(0.79)						
I (1995 < year < 2001)	−0.072**						
	(2.38)						
Real GDP growth	−0.011**						
	(2.20)						
Loan type fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	No	No	Yes	No	No	No	No
Other fixed effects	Industry	Firm	Firm	Industry	Industry	Industry	Industry
Observations	5,331	5,331	5,331	3,080	1,503	4,677	5,608
Adjusted R^2	0.69	0.81	0.83	0.69	0.71	0.68	0.68

5.9 Table 7: additional tariff-shock tests

Read:

- Adding macro controls leaves the coefficient at 0.154.
- With firm fixed effects, it is 0.183.
- With firm and year fixed effects, it falls to 0.080 but remains significant.
- In refinancing loans only, it is 0.245.
- In the subsample of firms observed borrowing both before and after the shock, it is 0.231.
- In the dynamic version, `Before1` and `Before0` are insignificant, while `After1`=0.220 and `After2+`=0.349.

- Import penetration also has a positive and significant effect.

Economic message. The tariff story survives macro controls, firm fixed effects, selection checks, and event-timing checks.

5.10 Table 8: change regressions

Read:

- The change in loan spreads is significantly related to the change in tariff rates in the expected direction.
- The effect remains after differencing firm characteristics.

Economic message. Even in a very demanding within-firm change design, falling tariffs go together with rising spreads.

Table 8
Changes in import tariff rates and changes in loan spreads.

This table presents coefficient estimates of the effect of changes in import tariff rates on loan spreads. The dependent variable is the change in the logarithm of loan spreads. *ΔTariff rate* is the change in the import tariff rate. For each firm that took out a loan of the same type before and after a significant tariff rate reduction, I identify the last loan before and the first loan after the tariff rate reduction. Based on these observations, I compute the change in the logarithm of loan spreads, tariff rates, and firm characteristics. The sample period is from 1992 to 2005. All variables are defined in Appendix A. I report *t*-statistics using standard errors adjusted for within-firm clustering in parentheses below the coefficient estimates. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

	(1)	(2)
ΔTariff rate	−0.141**	−0.117**
	(2.03)	(1.99)
ΔLog(total assets)		−0.213***
		(2.97)
ΔMarket-to-book		−0.083
		(1.56)
ΔLeverage		0.652
		(1.63)
ΔTangibility		−1.848
		(1.55)
ΔCash flow volatility		−0.854
		(1.55)
ΔDefault probability		1.472
		(0.49)
ΔProfitability		0.314**
		(1.99)
ΔLoan size		−0.280
		(0.91)
ΔLog(loan maturity)		−0.212**
		(2.22)
Observations	211	211
Adjusted R^2	0.02	0.18

Table 9
Cross-sectional variation in the effect of tariff rate reductions on loan spreads.

This table reports estimates of $Post\text{-}reduction_{j,t}$ from a series of estimations of Eq. (2). The dependent variable is the logarithm of loan spreads. $Post\text{-}reduction_{j,t}$ equals one if industry *j* has experienced a tariff rate reduction by time *t* that is larger than *three* times the median tariff rate reduction in that industry, and zero otherwise. I classify industries based on four variables: the HHI, the change in import penetration (*ΔImport penetration*), the average lagged advertising and R&D expense over sales, and the average lagged cash holdings to asset ratio of incumbent firms. For each year and for each proxy, I rank the sample industries according to their average value and assign firms from industries in the bottom and top quartiles to "low" and "high" industries, respectively. The standard errors for the differences between *High* and *Low* are computed with an SUR system that estimates the industry groups jointly. The last column *Diff. high–low* shows the *p*-value for the difference in the $Post\text{-}reduction_{j,t}$ coefficients between *High* and *Low* subgroups. All specifications include the control variables of the baseline specification (column 2, Table 6). I measure all independent variables as of the quarter prior to the loan start date. All estimations include loan type and industry fixed effects at the three-digit SIC code industry level. The sample period is from 1992 to 2005. All variables are defined in Appendix A. *t*-Statistics are in parentheses below the coefficient estimates and the number of observations in each group is in italics. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

	Low	*High*	*Diff. high–low*
Industry concentration and imports			
HHI	0.126***	0.416***	0.00***
	(2.93)	(9.34)	
	907	*797*	
ΔImport penetration	0.099**	0.253***	0.02**
	(1.97)	(5.53)	
	887	*754*	
Barriers to entry			
Advertising and R&D	0.316***	0.203***	0.08*
	(7.19)	(4.35)	
	878	*693*	
Cash holdings of incumbent firms	0.270***	0.148***	0.07*
	(6.00)	(2.98)	
	888	*683*	

5.11 Table 9: where tariff shocks matter most

Read:

- In competitive industries, the tariff-shock coefficient is 0.126; in concentrated industries, it jumps to 0.416.
- Where import penetration rises more, the coefficient is 0.253 versus 0.099 in low-penetration-change industries.
- The effect is larger when advertising and R&D barriers are low.
- The effect is also larger when incumbent cash holdings are low.

Economic message. The tariff design behaves exactly as it should if tariff cuts truly intensify competition. The shocks bite more where entry pressure is stronger and defensive barriers are weaker.

5.12 Table 10: financial strength and strategic interaction

Read:

- For small firms facing small rivals, the competition coefficient is essentially zero (0.007).
- For small firms facing large rivals, it is 0.151 in OLS and 0.209 in Fama–MacBeth.
- For unrated firms, the effect is larger when rivals have high credit ratings.
- In industries with high C4 concentration, the competition coefficient is 0.317 versus 0.089 in low-concentration industries.
- The similarity-of-operations split is somewhat weaker but still points to more competition risk where firms interact more intensely.

Economic message. Competition is particularly dangerous when the firm is strategically vulnerable to stronger rivals or to concentrated-product-market battles.

Table 10

Cross-sectional variation in the effect of competition on loan spreads: financial strength and strategic interactions.

This table reports estimates of *Competition* from a series of estimations of Eq. (1). The dependent variable is the logarithm of loan spreads. *Competition* is a dummy variable equal to one if the HHI at the three-digit SIC code industry level is in the lowest quartile of the yearly sample distribution, and zero otherwise. For *Rivals' size*, I classify *small* firms (below sample median) into two groups based on the median: firms facing small (*Low*) and large rivals (*High*). For *Rivals credit rating* I classify firms *without* a credit rating into two groups based on the average rivals' credit ratings: firms facing low (*Low*) and high (*High*) rated firms. For the *C4-Index* and the *Similarity of operations* proxy, each year and for each proxy, I rank the sample industries according to their average value and assign firms from industries in the bottom and top quartiles to "low" and "high" industries, respectively. The columns *Diff. high–low* show the *p*-value for the difference in the *Competition* coefficients between *High* and *Low* subgroups. All the estimations contain the control variables of the baseline specification (column 3, Table 3) and include loan type dummies. The OLS estimations include year and industry fixed effects at the three-digit SIC code industry level. The sample period is from 1992 to 2007. All variables are defined in Appendix A. *t*-Statistics are in parentheses below the coefficient estimates and the number of observations in each group is in italics. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

	OLS			*Fama and MacBeth*		
	Low	*High*	*Diff. high–low*	*Low*	*High*	*Diff. high–low*
Differences in financial strength						
Rivals' size for small firms	−0.007	0.151***	0.00***	−0.007	0.209***	0.00***
	(0.19)	(3.42)		(0.21)	(3.83)	
	2,971	*2,968*		*2,971*	*2,968*	
Rivals' rating for unrated firms	0.061*	0.087*	0.67	0.022	0.142**	0.04**
	(1.72)	(1.75)		(0.97)	(2.16)	
	4,512	*2,884*		*4,512*	*2,884*	
Strategic interactions						
Industry concentration (C4-Index)	0.089**	0.317***	0.00***	−0.001	0.140**	0.03**
	(1.96)	(4.21)		(0.04)	(2.46)	
	3,001	*2,938*		*3,001*	*2,938*	
Similarity of operations	0.166***	0.139***	0.73	0.069	0.155***	0.22
	(2.97)	(2.56)		(1.45)	(2.98)	
	2,703	*2,753*		*2,703*	*2,753*	

Table 11

Cross-sectional variation in the effect of competition on loan spreads: asset specificity and illiquidity.

This table reports estimates of *Competition* from a series of estimations of Eq. (1). The dependent variable is the logarithm of loan spreads. *Competition* is a dummy variable equal to one if the HHI at the three-digit SIC code industry level is in the lowest quartile of the yearly sample distribution, and zero otherwise. I classify industries based on four variables: the proportion of machinery and equipment to total assets (*Asset specificity*), the inverse of the quick ratio (*Asset illiquidity*), the average leverage net of cash of rival firms (*Net leverage of rivals*), and the number of competitors in the same industry with a credit rating (*Number of potential buyers*). For each year and for each proxy, I rank the sample industries according to their average value and assign firms from industries in the bottom and top quartiles to "low" and "high" industries, respectively. The columns *Diff. high–low* show the *p*-value for the difference in the *Competition* coefficients between *High* and *Low* subgroups. All the estimations contain the control variables of the baseline specification (column 3, Table 3) and include loan type dummies. The OLS estimations include year and industry fixed effects at the three-digit SIC code industry level. The sample period is from 1992 to 2007. All variables are defined in Appendix A. *t*-Statistics are in parentheses below the coefficient estimates and the number of observations in each group is in italics. Significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

	OLS			*Fama and MacBeth*		
	Low	*High*	*Diff. high–low*	*Low*	*High*	*Diff. high–low*
Asset specificity	0.106**	0.289***	0.03**	−0.015	0.0584	0.40
	(2.11)	(4.04)		(0.45)	(0.72)	
	2,901	*2,712*		*2,901*	*2,712*	
Asset illiquidity	0.096**	0.243***	0.11	−0.004	0.239***	0.00***
	(2.28)	(2.91)		(0.13)	(3.94)	
	3,155	*2,957*		*3,155*	*2,957*	
Net leverage of rivals	0.103***	0.206***	0.09*	0.001	0.138***	0.01**
	(2.82)	(4.11)		(0.04)	(2.93)	
	3,174	*2,816*		*3,174*	*2,816*	
Number of potential buyers	0.189***	0.109**	0.28	0.116*	−0.011	0.10*
	(3.06)	(2.66)		(1.80)	(0.18)	
	3,182	*2,960*		*3,182*	*2,960*	

5.13 Table 11: asset specificity and asset illiquidity

Read:

- With high asset specificity, the competition coefficient is 0.289 versus 0.106 in low-specificity industries.
- With high asset illiquidity, it is 0.243 versus 0.096 in the OLS split.
- When rivals have high net leverage, the competition effect is 0.206 versus 0.103.
- When the number of potential buyers is small, the effect is larger than when many buyers exist.

Economic message. Competition is more costly for debt when collateral is harder to liquidate. This is the loss-given-default channel showing up in the data.

6 Short summary

This paper shows that product market competition raises firms' bank loan spreads. The baseline panel evidence says that firms in the most competitive industries pay materially higher spreads even after controlling for loan characteristics, firm risk, ratings, lender effects, and governance. The core causal test uses large import tariff reductions as quasi-exogenous increases in foreign competitive pressure and finds that spreads rise after these shocks.

The paper also shows that the effect is not uniform. It is strongest when weak firms face strong rivals, when industries are strategically intense, and when assets are specific or illiquid. These patterns suggest that banks price not only conventional default risk but also broader strategic and liquidation risk generated by product market competition.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes a clear empirical link between competition in the product market and the price of bank debt. More competition means higher loan spreads.

7.2 Economic intuition

Banks care about whether the borrower will survive and what can be recovered if the borrower fails. Competition worsens both objects. It can make cash flows more fragile, intensify predatory pressure from rivals, and reduce the resale value of assets in distress. Because of that, banks charge more.

7.3 Takeaway

The paper's deepest message is that product-market structure belongs inside corporate finance. Debt pricing is not determined only by balance-sheet ratios and standard credit scores. It is also shaped by who a firm's rivals are, how aggressively they can compete, and how liquid the firm's assets would be in distress.